

A Real-time Research Project/Societal Related Project

Report on

AI-Powered Quiz Generator – Quizzzy

Submitted in Partial fulfillment of requirements for B.Tech II Year II Semester course

By

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KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY

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Narayanaguda, Hyderabad, Telangana-29

2025-26



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CERTIFICATE

This is to certify that this is a bonafide record of the project report titled “**AI-Powered Quiz Generator – Quizzzy**” which is being presented as the **Real-time Research Project / Societal Related Project** report by

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In partial fulfillment for the II Year II Semester Course RTRP in KMIT affiliated to the Jawaharlal Nehru Technological University, Hyderabad

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Submitted for Final Project Review held on _____

Vision & Mission of KMIT

Vision

- To be the fountain head in producing highly skilled, globally competent engineers.
- Producing quality graduates trained in the latest software technologies and related tools and striving to make India a world leader in software products and services.

Mission

- To provide a learning environment that inculcates problem solving skills, professional, ethical responsibilities, lifelong learning through multi model platforms and prepares students to become successful professionals.
- To establish an industry institute Interaction to make students ready for the industry.
- To provide exposure to students on the latest hard ware and software tools.
- To promote-research based projects/activities in the emerging areas of technology convergence.
- To encourage and enable students to not merely seek jobs from the industry but also to create new enterprises.
- To induce a spirit of nationalism which will enable the student to develop, understand India's challenges and to encourage them to develop effective solutions.
- To support the faculty to accelerate their learning curve to deliver excellent service to students.

PROGRAM OUTCOMES(POs)

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PO3. Design/Development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

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PO5. Modern Tool Usage: Create select, and, apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The Engineer and Society: Apply reasoning informed by contextual knowledge to societal, health, safety. Legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi disciplinary environments.

PO12. Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



PROJECT OUTCOMES

P1: To develop an AI-powered quiz generation system using modern web technologies.

P2: To automate quiz creation from topics, PDF documents, and web URLs using AI and NLP techniques.

P3: To design an interactive and user-friendly platform with timer-based quizzes, scoreboards, and gamification features.

P4: To implement a scalable and efficient client-server application for real-time quiz generation and evaluation.

MAPPING PROJECT OUTCOMES WITH PROGRAM OUTCOMES

PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
P1	H	M	H	M	H	L	L	L	M	M	L	M
P2	M	H	H	M	H	L	L	L	M	M	M	H
P3	M	H	M	H	M	L	L	L	M	H	M	M
P4	L	M	M	M	M	L	L	L	H	M	L	M

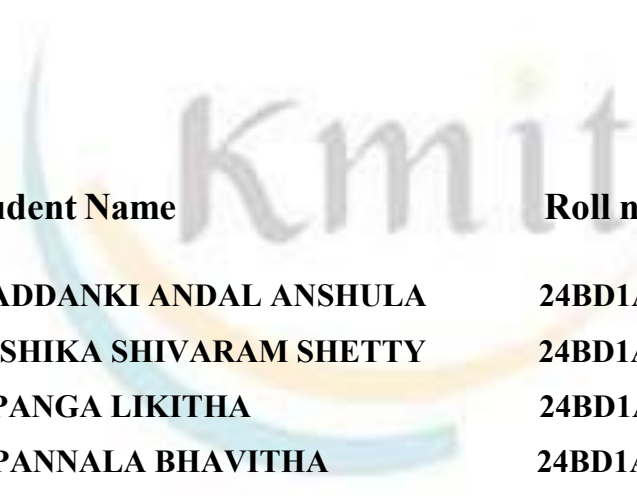
L– LOW

M –MEDIUM

H– HIGH

DECLARATION

We hereby declare that the results embodied in the dissertation entitled “**AI Powered Quiz Generator--Quizzy**” has been carried out by us together during the academic year 2025-26 as a partial fulfillment of the B.Tech II Year II Semester Course “**Real-time Research Project / Societal Related Project**”. We have not submitted this report to any other Course/College.



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ACKNOWLEDGEMENT

We take this opportunity to thank all the people who have rendered their full support to our project work. We render our thanks to **Dr. B L Malleswari**, Principal who encouraged us to do the Project.

We are grateful to **Mr. Neil Gogte**, Founder & Director and **Mr. S. Nitin**, Director, for facilitating all the amenities required for carrying out this project.

We express our sincere gratitude to **Ms. Deepa Ganu**, Academic Director for providing an excellent environment in the college.

We are also thankful to **Mr. Shailesh Gangakhedkar**, Real-Time Research Project Program Coordinator for providing us with time to make this project a success within the given schedule.

We are also thankful to our Project Mentor **Ms. G. Swetha Goud** for her/his valuable guidance and encouragement given to us throughout the project work.

We would like to thank the entire KMIT faculty, who helped us directly and indirectly in the completion of the project.

We sincerely thank our friends and family for their constant motivation during the project work.

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ABSTRACT

Quizzzy is an AI-powered web application that automates quiz generation from topics, PDF documents, and web URLs. It helps reduce the manual effort required to create quizzes and provides a personalized learning experience for students.

The system allows users to customize quizzes by selecting the number of questions, timer settings, question type, and difficulty level. It also supports adaptive difficulty, instant evaluation, AI-generated explanations, flashcards, leaderboard, dashboard analytics, teacher dashboard, study recommendations, and a Brain Lab module for interactive practice.

Quizzzy is developed using HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, and AI API integration. The project demonstrates how Artificial Intelligence can be used in education to improve quiz creation, learning assessment, and student engagement.

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CHAPTER-1

1. INTRODUCTION

Quizzzy is an AI-powered quiz generation platform designed to automate the process of creating quizzes using Artificial Intelligence and Natural Language Processing. Traditional quiz creation requires manual effort and can be time-consuming, especially when questions need to be prepared from different sources. Quizzzy solves this problem by allowing users to generate quizzes from topics, PDF documents, and web URLs.

The system supports customizable quiz generation, including number of questions, timer settings, question type, and difficulty level. It also provides adaptive quiz options, instant evaluation, AI-generated explanations, flashcards, leaderboard, dashboard analytics, profile-based progress tracking, teacher dashboard, study recommendations, and Brain Lab activities for interactive practice.

Quizzzy supports different user types such as students, teachers, and self-learners. Users can register and login using email, parent phone number, or roll number/unique ID. The platform aims to improve learning efficiency, reduce manual quiz preparation, and provide an engaging self-learning and assessment experience.

1.1 Purpose of the Project

The purpose of this project is to develop an AI-powered quiz generation system that reduces the manual effort required to create quizzes. Quizzzy allows users to generate quizzes from topics, PDF documents, and web URLs, making the assessment process faster and more flexible.

The project also aims to provide a personalized learning experience through customizable quiz settings, adaptive difficulty, instant evaluation, AI-generated explanations, flashcards, dashboard analytics, leaderboard, teacher dashboard, and study recommendations. This helps students practice more effectively and helps teachers monitor real student performance in a structured way..

1.2 Problem with Existing Systems

Existing quiz systems usually require teachers or users to manually prepare questions, options, answers, and explanations. This process is time-consuming and becomes difficult when quizzes need to be created from large documents, online resources, or specific topics.

Many existing systems also lack personalization features such as adaptive difficulty, performance tracking, study recommendations, and detailed explanations. As a result, students may receive quiz scores but do not always get enough guidance to understand mistakes or improve weak areas. This creates the need for an AI-based system that can generate quizzes automatically and support personalized learning.

1.3 Proposed System

The proposed system, Quizzy, is an AI-powered quiz generation platform that allows users to create quizzes from topics, PDF documents, and web URLs. The system uses Artificial Intelligence and Natural Language Processing techniques to understand the input content and generate relevant questions, options, answers, and explanations.

Quizzy provides customizable quiz settings such as number of questions, timer options, question type, and difficulty level. It also supports adaptive quiz functionality, where the difficulty can change during the quiz based on the user's answers. The system includes instant evaluation, result analysis, flashcards, leaderboard, dashboard analytics, profile-based progress tracking, teacher dashboard, study recommendations, and Brain Lab activities to make learning more interactive and personalized.

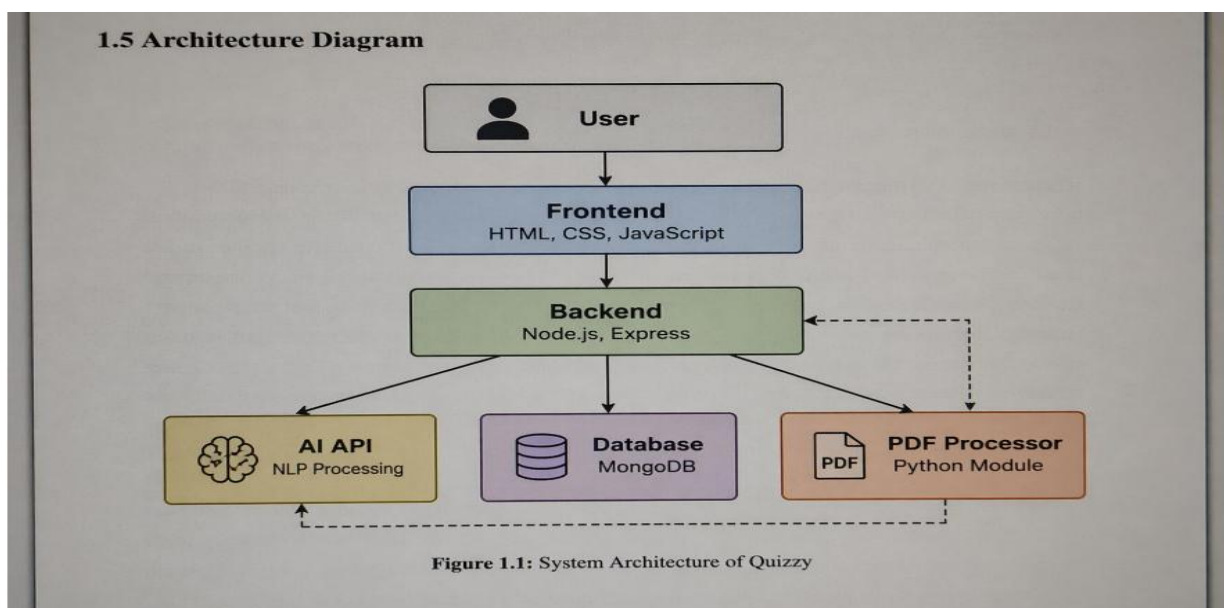
The system supports different account types such as Student, Teacher, and Self Learner. Users can register and login using email, parent phone number, or roll number/unique ID. Teachers can view real student quiz attempts, accuracy, topic performance, and recent activity through the teacher dashboard.

1.4 Scope of the Project

The scope of Quizzy includes AI-based quiz generation, automatic evaluation, personalized learning support, and progress tracking. Users can generate quizzes from topics, PDF documents, and URLs, select quiz settings, attempt quizzes, view results, and revise using flashcards.

The system also provides dashboard analytics, leaderboard ranking, profile progress, teacher dashboard, study recommendations, and Brain Lab activities for engagement. The project focuses on helping students practice effectively, understand mistakes through explanations, and improve learning outcomes through technology-based assessment

1.5 Architecture Diagram



CHAPTER-2

2. LITERATURE SURVEY

Various online quiz platforms such as Google Forms and Kahoot provide tools for quiz creation and participation. These platforms allow users to create and conduct quizzes efficiently; however, they require manual input of questions and do not support automated quiz generation.

With the advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), automated content generation has become possible. AI-based systems can analyze textual data and generate meaningful questions from it. These technologies are increasingly being used in educational applications to improve learning efficiency and reduce manual effort.

Some existing systems provide partial automation but lack support for multiple input formats such as PDF documents and web URLs. Additionally, most platforms do not include real-time evaluation or engaging features such as gamification, which are important for improving user interaction and learning outcomes.

The proposed system, Quizzy, overcomes these limitations by integrating AI-based quiz generation with support for multiple input sources. It also enhances user engagement through features such as a dynamic scoreboard and Brain Lab games. This makes the system more efficient, flexible, and interactive compared to traditional quiz platforms.

CHAPTER-3

3. SOFTWARE REQUIREMENTS SPECIFICATION

3.1 Introduction to SRS

The Software Requirements Specification (SRS) defines the functional and non-functional requirements of the Quizzzy system. It serves as a guideline for developers, testers, and stakeholders to understand the system behavior and expected outcomes.

This document provides a clear description of system features, constraints, and performance requirements. It ensures that the system is developed according to user needs and project objectives.

3.2 Role of SRS

The SRS plays a crucial role in the development process by providing a detailed description of system requirements. It helps in planning, designing, and implementing the system effectively.

It also serves as a reference for testing and validation, ensuring that the developed system meets all specified requirements. The SRS helps maintain consistency and reduces ambiguity in system development.

3.3 Requirements Specification Document

The Quizzzy system is an AI-based web application designed to automate quiz generation. The system allows users to generate quizzes from topics, PDF documents, and web URLs.

It supports multiple question formats such as MCQ and short answer questions. The system evaluates answers automatically and provides results through a scoreboard. It also includes a Brain Lab module to enhance user engagement.

3.4 Functional Requirements

- The system shall allow users to register as Student, Teacher, or Self Learner.
- The system shall allow users to login using email, parent phone number, or roll number/unique ID.
- The system shall allow users to generate quizzes from topics, PDF documents, and web URLs.
- The system shall allow users to select the number of questions, with a minimum of 5 questions.
- The system shall allow users to enable or disable the quiz timer.
- If the timer is enabled, the system shall allow users to set seconds per question with a minimum of 15 seconds.
- The system shall allow users to select difficulty levels such as Easy, Medium, Hard, and Adaptive.

- The system shall generate quiz questions, answer options, correct answers, and explanations using AI.
- The system shall evaluate user answers automatically and display the score.
- The system shall display explanations and recommendations after quiz completion.
- The system shall store quiz history and show progress through the dashboard and profile page.
- The system shall provide a teacher dashboard to view real student quiz attempts, accuracy, topic performance, and recent activity.
- The system shall provide flashcards for revision.
- The system shall display leaderboard rankings for user engagement.
- The system shall provide Brain Lab activities for interactive cognitive practice.

3.5 Non-Functional Requirements

- The system should provide a responsive and user-friendly interface.
- The system should generate quizzes within a reasonable time based on the number of questions and input size.
- The system should maintain readability and accessibility in both light and dark modes.
- The system should protect user data through authentication and secure session handling.
- The system should allow flexible login using email, parent phone number, or roll number/unique ID.
- The system should be scalable enough to support multiple users, student progress tracking, and teacher dashboard access.
- The system should handle invalid inputs, unsupported files, and API errors gracefully.

3.6 Performance Requirements

- The system should process topic-based quiz generation quickly for normal question counts.
- The system should support larger quizzes by generating questions in a controlled manner without overloading the AI API.
- The system should provide clear validation messages for invalid question count and timer inputs.
- The dashboard, profile, flashcards, leaderboard, teacher dashboard, and Brain Lab pages should load efficiently and display stored user data correctly.

3.7 Software Requirements

Frontend Technologies:

- HTML
- CSS
- JavaScript

Backend Technologies:

- Node.js
- Express.js

Database:

- MongoDB

Other Tools:

- AI API for quiz generation
- Python libraries for PDF processing

3.8 Hardware Requirements

The system can be accessed using standard computing devices such as desktop computers, laptops, tablets, and smartphones. A stable internet connection is required for system functionality

CHAPTER-4

4. SYSTEM DESIGN

4.1 Introduction to UML

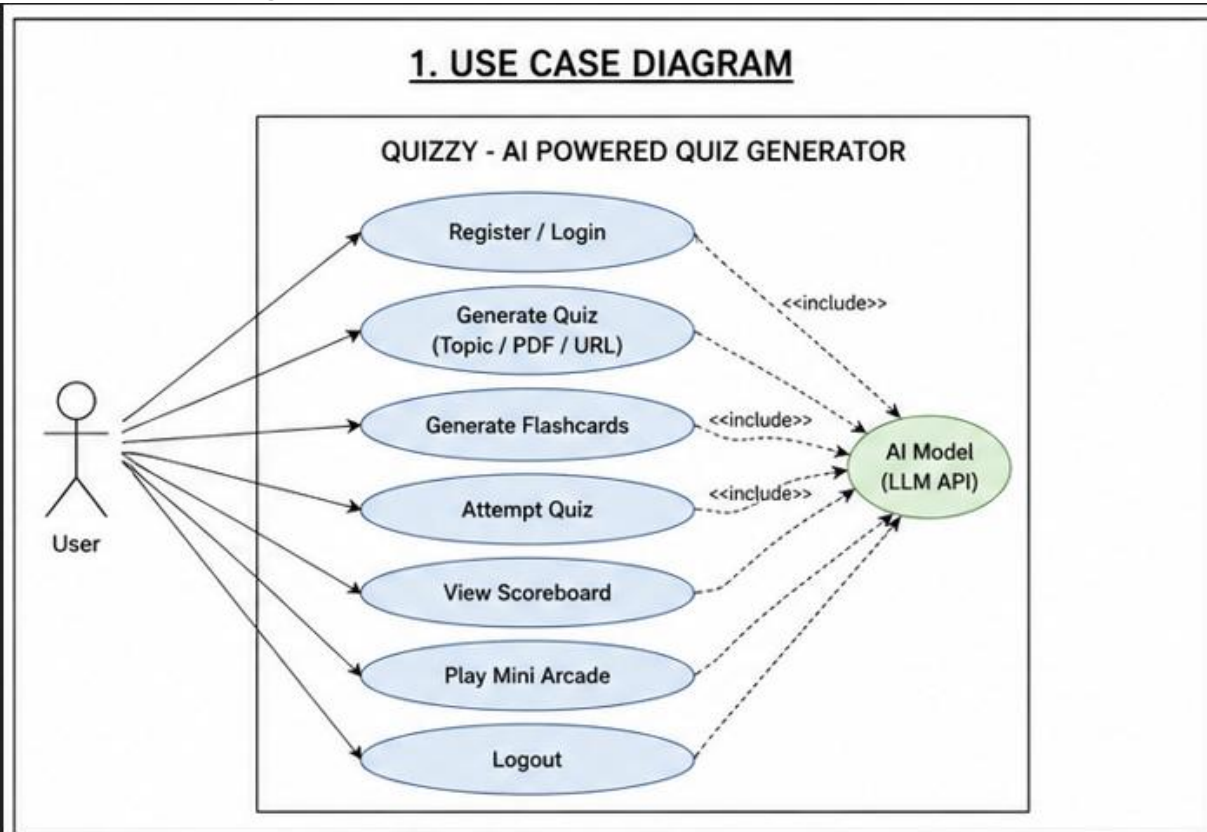
Unified Modeling Language (UML) is used to represent the design and structure of the system. It provides a visual representation of system components, interactions, and workflows. In the Quizzzy system, UML diagrams are used to illustrate user interactions, system processes, and the overall architecture. These diagrams help in understanding how different components of the system work together.

4.2 UML Diagrams

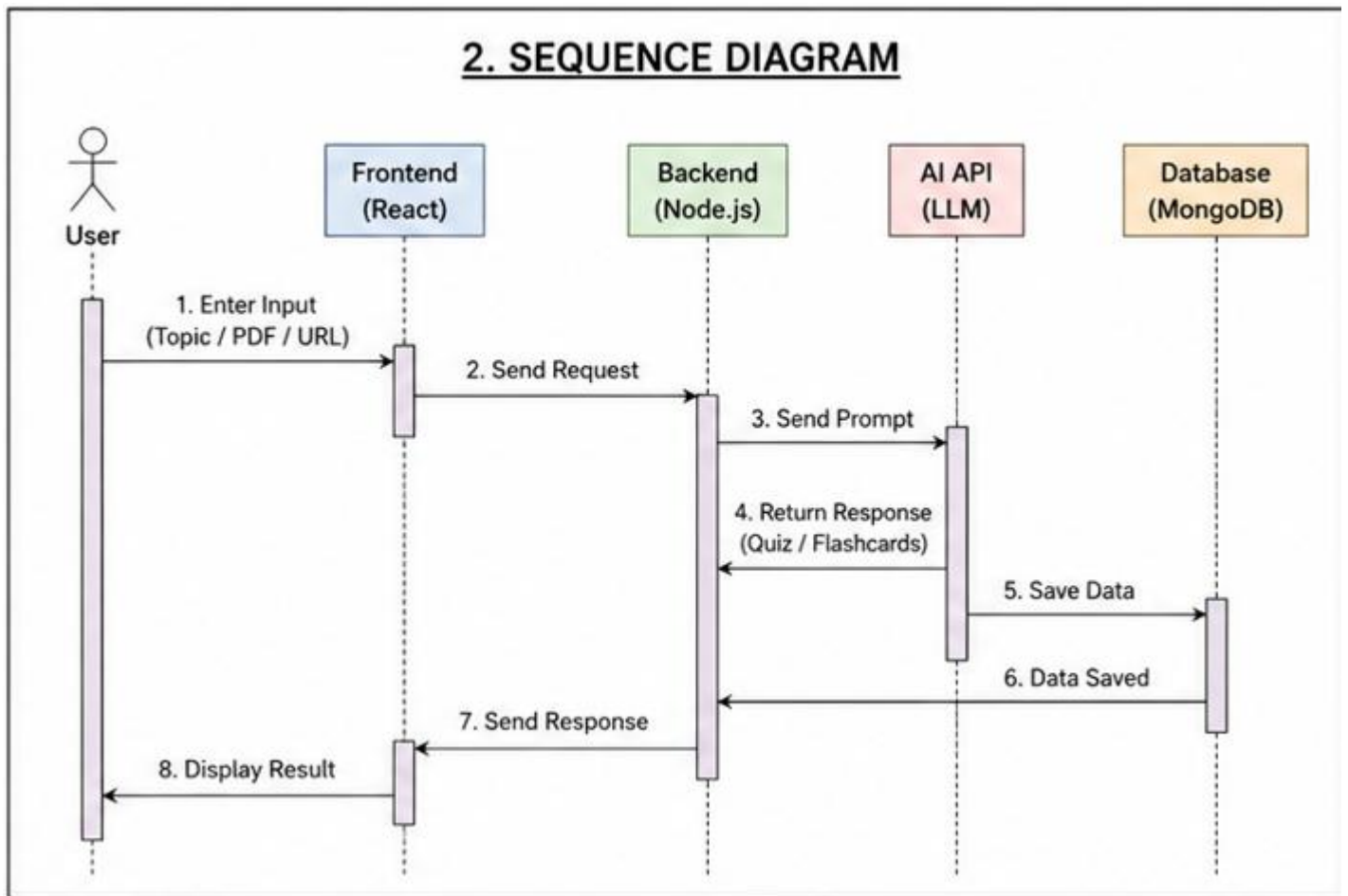
The following UML diagrams are used to represent the system design of Quizzzy:

- Use Case Diagram
- Sequence Diagram
- State Chart Diagram
- Deployment Diagram

4.2.1 Use Case Diagram



4.2.2 Sequence Diagram



4.2.3 State Chart Diagram

STATE CHART DIAGRAM

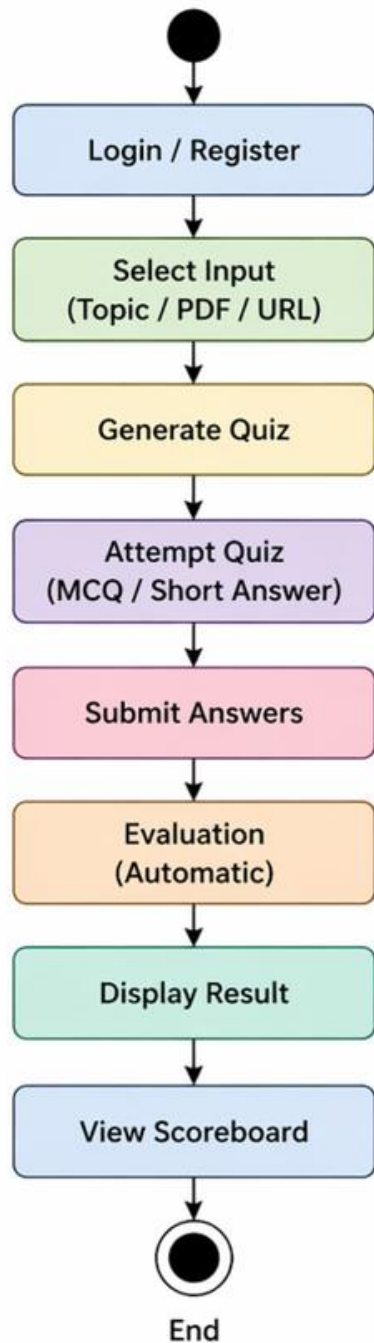


Figure 4.2.3: State Chart Diagram

4.2.4 Deployment Diagram

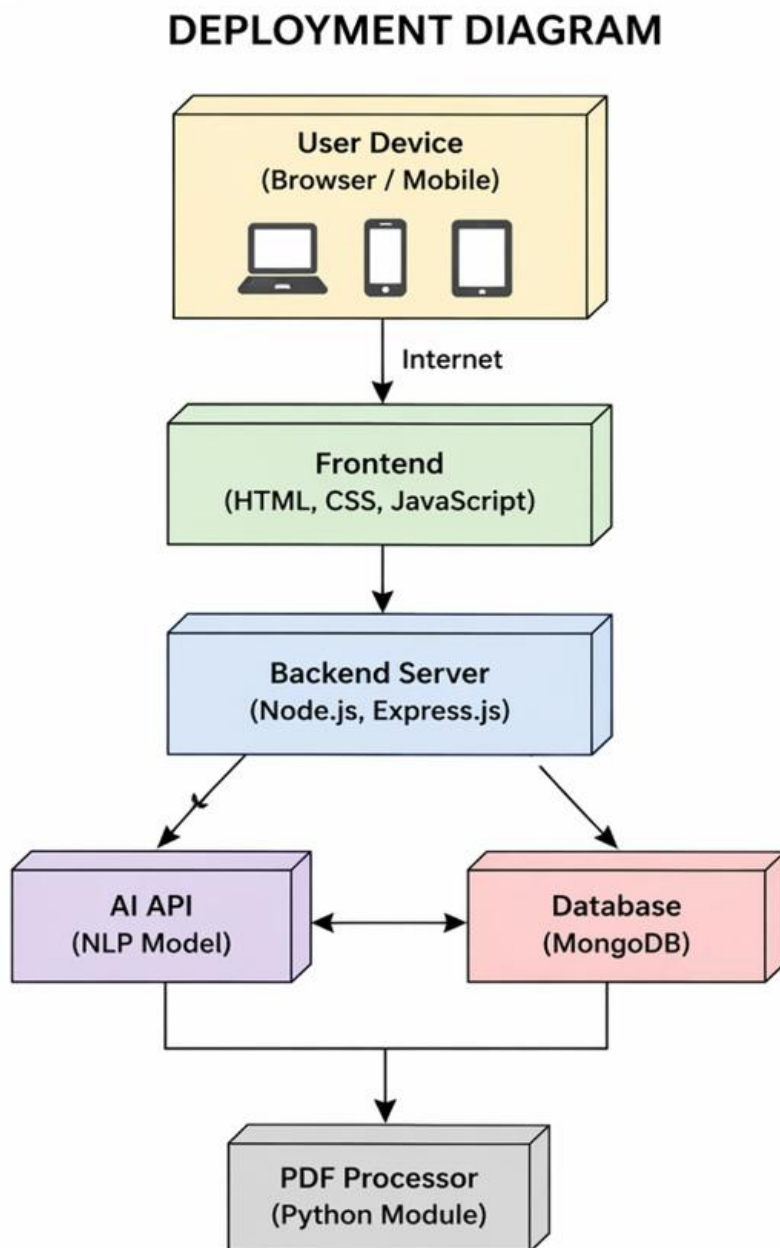


Figure 4.2.4: Deployment Diagram

4.3 Technologies Used

Frontend technologies used in Quizzzy include HTML, CSS, JavaScript, and Vite. These are used to build responsive pages such as the homepage, quiz generation page, quiz interface, results page, dashboard, flashcards, profile, leaderboard, teacher dashboard, and Brain Lab.

Backend technologies include Node.js and Express.js for handling API requests, quiz generation, authentication, content extraction, teacher dashboard data, and evaluation. MongoDB is used for storing user data, quiz history, leaderboard data, flashcards, and progress-related information.

The system uses AI API integration for generating quiz questions, answer options, explanations, and learning recommendations. Additional libraries are used for PDF parsing, URL content extraction, charts, icons, authentication, and secure data handling.

CHAPTER-5

5. IMPLEMENTATION

Quizzzy is implemented using a client-server architecture. The frontend provides interactive pages for quiz generation, quiz attempts, results, dashboard, profile, flashcards, leaderboard, and Brain Lab. The backend handles authentication, quiz generation requests, content extraction, AI API communication, answer evaluation, and data storage.

When a user provides a topic, PDF, or URL, the system processes the input and sends the required information to the backend. The AI model generates questions according to the selected number of questions, difficulty, question type, and timer settings. The generated quiz is then displayed to the user through the quiz interface.

After the quiz is submitted, the system evaluates the answers, calculates the score, displays explanations, and stores the attempt in the user history. The dashboard and profile pages use this history to show performance analytics, progress, leaderboard position, flashcards, study recommendations, and performance prediction. The Brain Lab module provides additional interactive activities for memory, reaction, and recall practice.

The application is deployed as a web-based system and can be accessed from different devices through a browser.

5.1 Setting up connections

This section describes the setup and integration of different components of the Quizzzy system. The frontend is developed using HTML, CSS, and JavaScript, and it communicates with the backend through HTTP requests.

The backend is built using Node.js and Express.js and is responsible for handling user inputs and processing quiz generation. It is connected to a MongoDB database to store user data and quiz results.

The backend also integrates with an AI API to generate quiz questions dynamically. For PDF based input, a Python module is used to extract text from uploaded documents before processing it through the AI system.

5.2 Coding the logic

This section explains the implementation logic of the system. When a user provides input in the form of a topic, PDF, or URL, the backend processes the input and sends it to the AI API. The AI system analyzes the input using Natural Language Processing techniques and generates relevant quiz questions. The system supports multiple formats such as MCQs and

short answer questions.

A timer mechanism is implemented for MCQs to simulate real-time testing. After the user submits answers, the system evaluates the responses automatically and calculates the score based on correctness.

5.3 Connecting the dashboard

This section describes the user interface and result display components of the system. The frontend displays quiz questions, collects user responses, and sends them to the backend for evaluation.

After evaluation, results are displayed to the user along with a dynamic scoreboard. The scoreboard ranks users based on their performance.

The interface is designed to be responsive and user-friendly. A global styling system is used to maintain consistency across all pages. A theme toggle feature is also implemented to switch between dark and light modes.

5.4 Screenshots

Figure 5.4.1: Homepage of Quizzzy

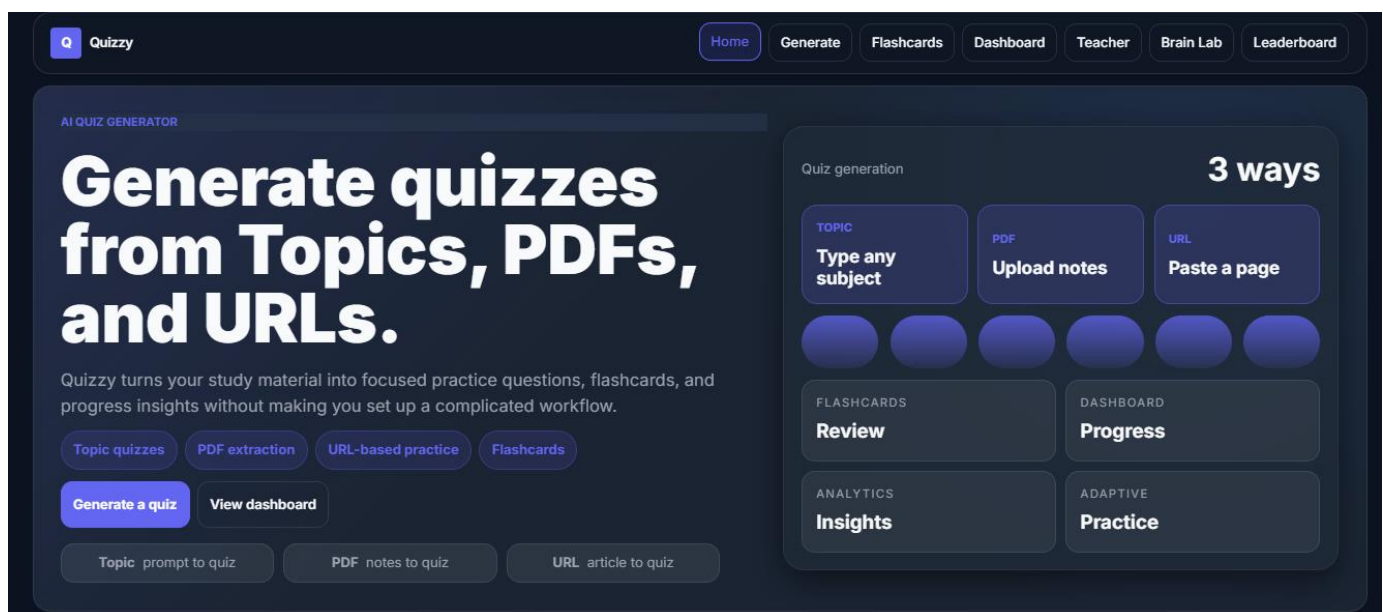


Figure 5.4.2: Quiz Generation Page

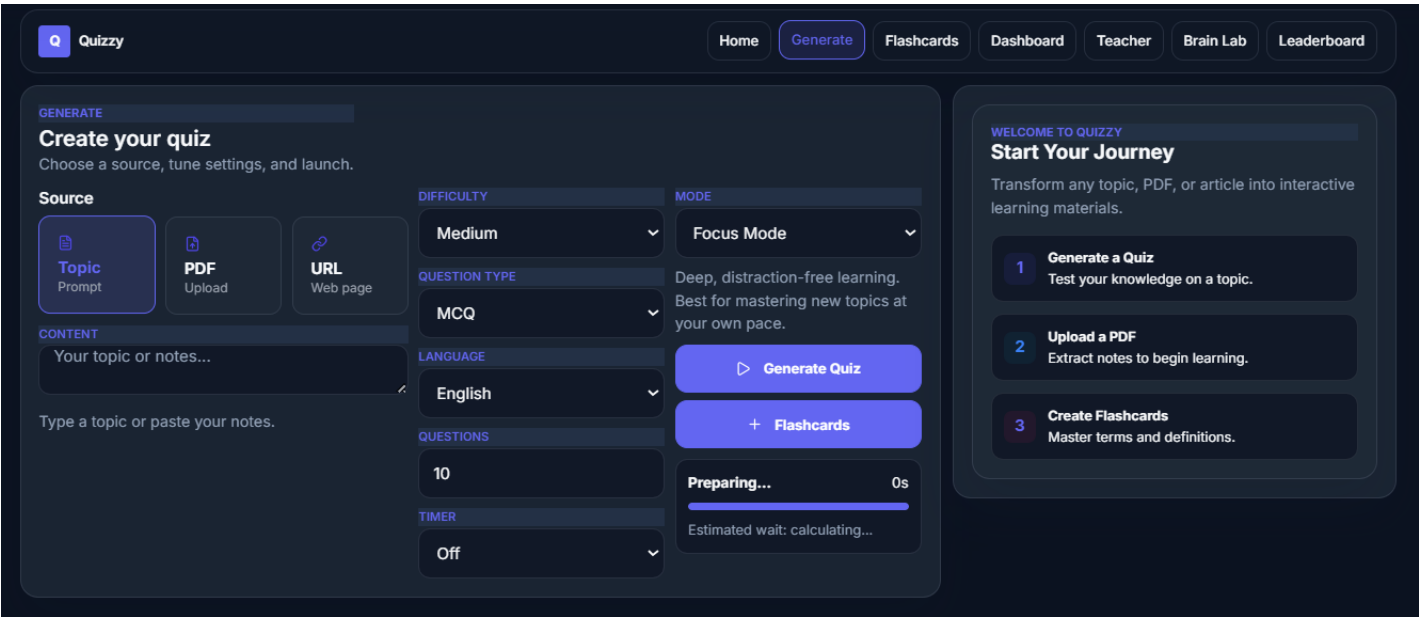


Figure 5.4.3: Quiz Interface Page

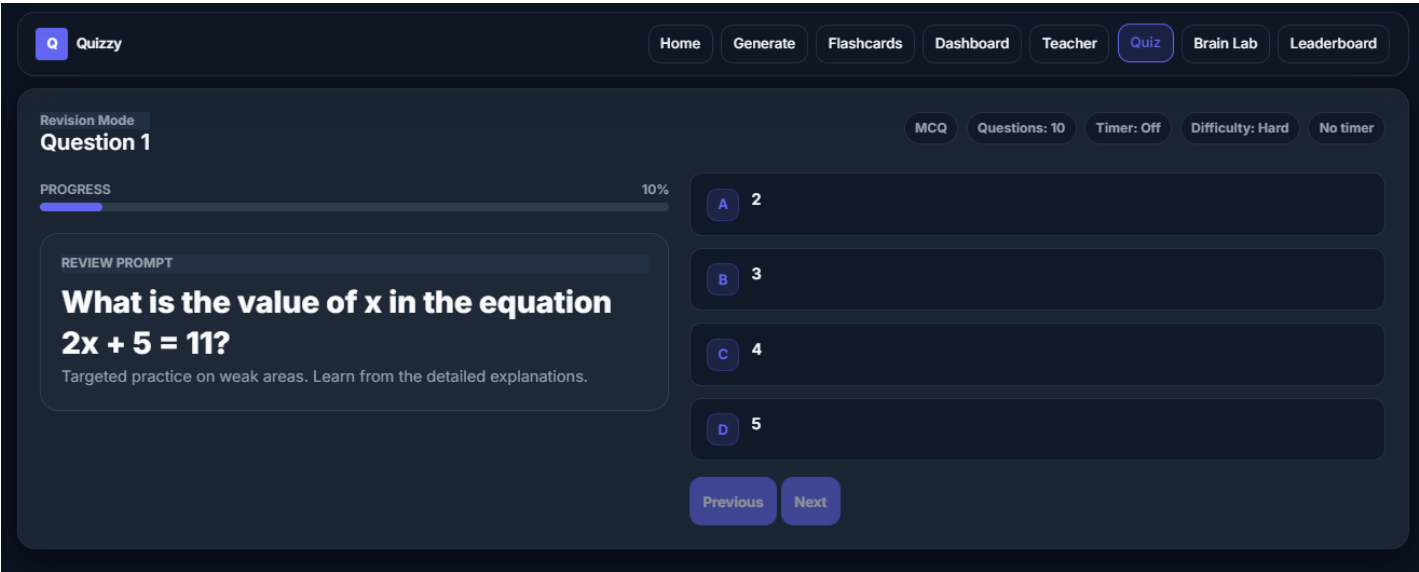


Figure 5.4.4: Quiz Result and Recommendation Page

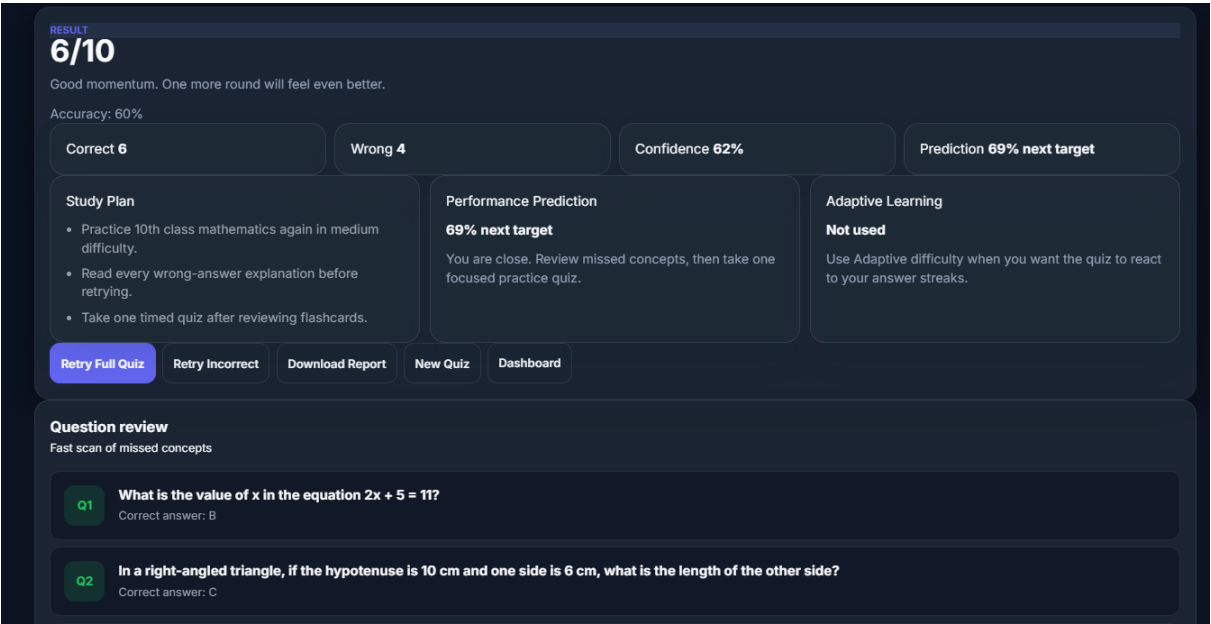


Figure 5.4.5: Dashboard & Analytics Page

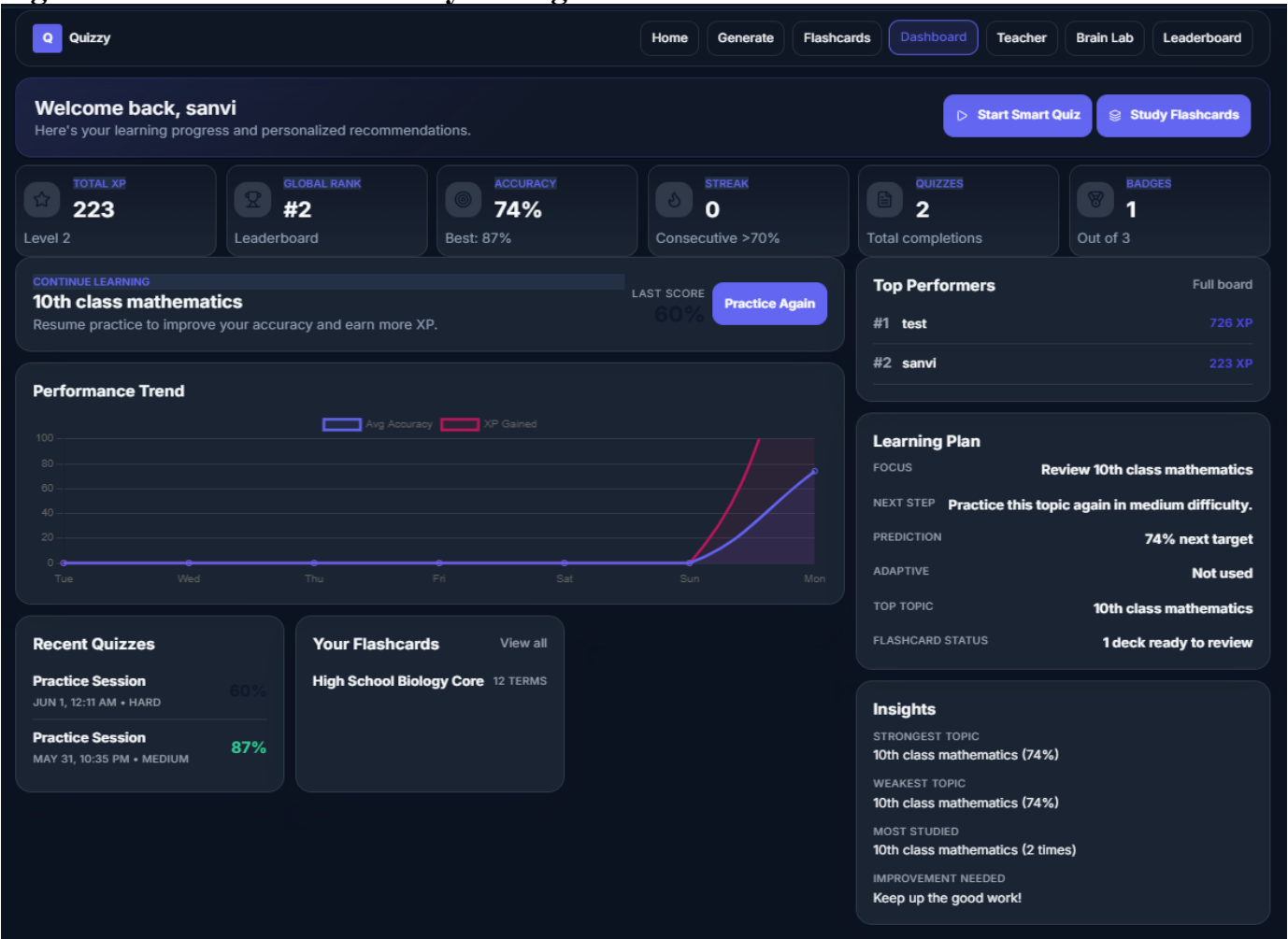


Figure 5.4.6:Teacher Dashboard

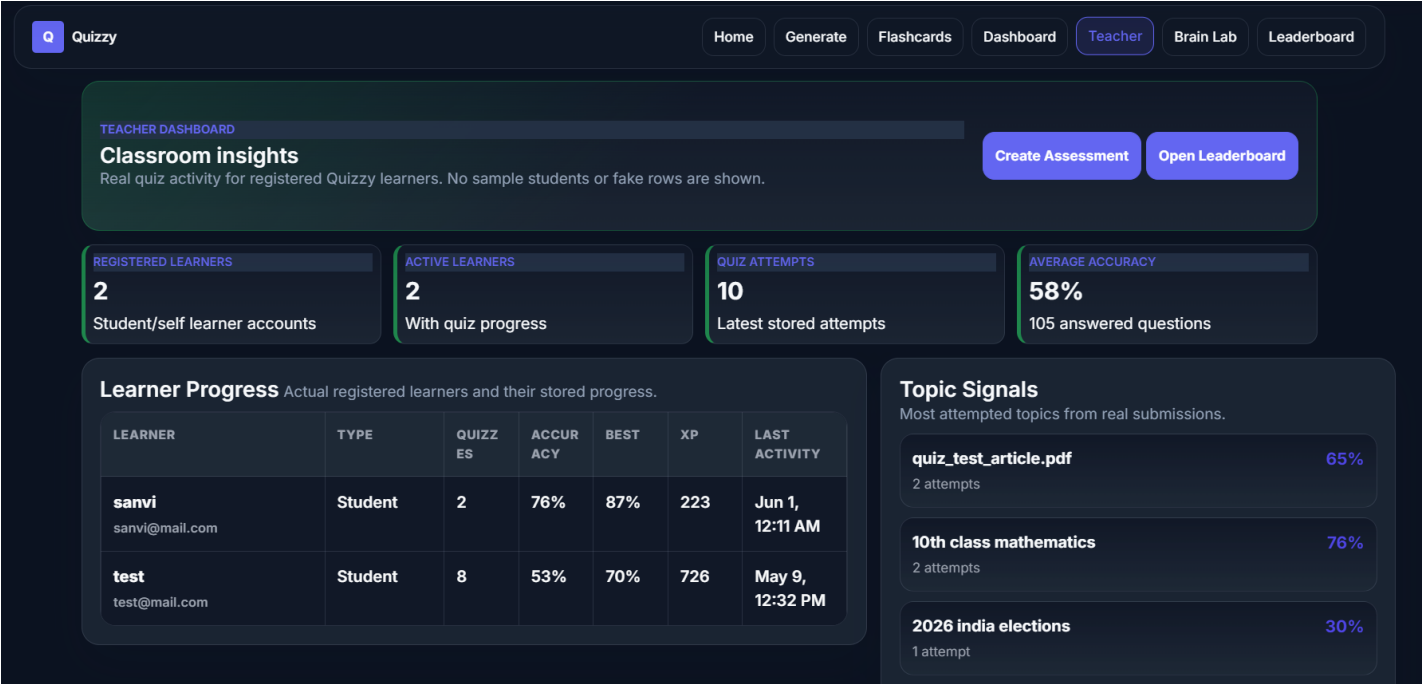
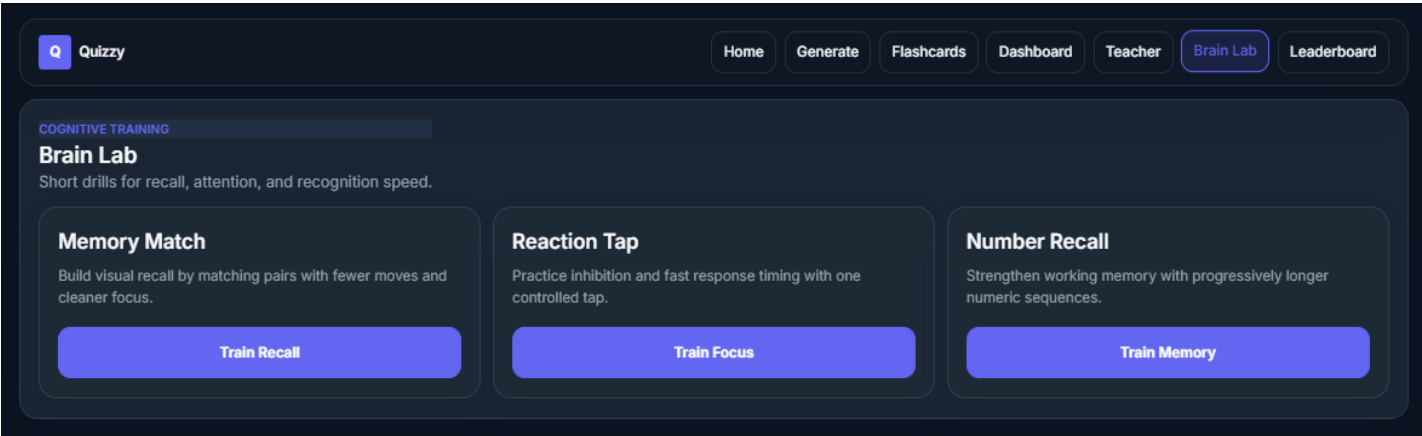


Figure 5.4.7: Brain Lab



CHAPTER-6

6.TESTING

6.1 Introduction

Testing is an essential phase in software development used to verify that the system functions correctly and meets the specified requirements. The Quizzy system was tested to ensure accurate quiz generation, correct evaluation of answers, and smooth user interaction.

Various testing methods were applied to validate system functionality, performance, and reliability.

6.1.1 Testing Objectives

The main objectives of testing the Quizzy system are:

- To ensure quiz generation works correctly for all input types
- To verify correct evaluation of user responses
- To check the functionality of timer-based MCQs
- To ensure the scoreboard displays accurate results
- To validate the Brain Lab module functionality
- To ensure smooth and error-free user interaction

6.1.2 Testing Strategies

The following testing strategies were used:

- Functional Testing: To verify all features such as quiz generation, evaluation, and scoreboard
- Unit Testing: To test individual components of the system
- Integration Testing: To ensure proper communication between frontend, backend, and AI API
- User Interface Testing: To verify usability and responsiveness of the system

6.1.3 System Evaluation

The system was evaluated based on performance, usability, and reliability.

The Quizzy system generates quizzes within a few seconds and handles user interactions efficiently. The user interface is simple, intuitive, and responsive across different devices.

The system operates without crashes and provides accurate results, demonstrating reliability and stability.

6.1.4 Testing New System

The developed system was tested with different inputs such as topics, PDF documents, and URLs. All functionalities were verified under various conditions.

The system successfully generated quizzes, evaluated answers correctly, and displayed results without errors. The Brain Lab module was also tested and functioned as expected.

6.2 Test Cases

Test Case	Description	Expected Result	Actual Result
TC1	Generate quiz using topic	Quiz generated successfully	Passed
TC2	Generate quiz using PDF	Quiz generated from document	Passed
TC3	Generate quiz using URL	Quiz generated from URL	Passed
TC4	Attempt quiz	Questions displayed correctly	Passed
TC5	Submit answers	Correct evaluation shown	Passed
TC6	View scoreboard	Performance displayed correctly	Passed
TC7	View flashcards	Flashcards displayed correctly	Passed
TC8	Access Brain Lab	Games load and run	Passed

CONCLUSION

Quizzy successfully demonstrates how Artificial Intelligence can be used to automate quiz creation and improve the learning experience. The system allows users to generate quizzes from topics, PDF documents, and web URLs, reducing manual effort and making practice more flexible.

With features such as customizable quiz settings, adaptive difficulty, instant evaluation, AI-generated explanations, flashcards, dashboard analytics, teacher dashboard, leaderboard, profile tracking, study recommendations, and Brain Lab activities, Quizzy provides a complete and interactive learning platform. The project shows the practical use of AI in education and supports personalized, self-paced learning and teacher-assisted progress monitoring.

FUTURE ENHANCEMENTS

- In the future, Quizzzy can be improved by developing an advanced adaptive learning engine that analyzes long-term user performance and adjusts quiz difficulty more intelligently.
- Classroom management tools can be introduced to organize students into classes, manage assignments, and track learning outcomes more deeply.
- Collaborative quiz sessions can be added so multiple users can participate in live quizzes together.
- A mobile application can be developed to make Quizzzy more accessible on smartphones and tablets.
- Advanced performance prediction analytics can be added to provide deeper insights into student strengths, weaknesses, and expected improvement.
- Parent progress monitoring can be added so parents can view student performance and learning progress.
- Assignment creation and submission workflow can be added to help teachers assign quizzes and review student submissions.

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5. Express.js Official Documentation. Available: <https://expressjs.com>
6. OpenAI API Documentation. Available: <https://platform.openai.com/docs>
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